

A Lagrangian Conditional Gaussian Koopman Network (LaCGKN) for Data Assimilation and Prediction

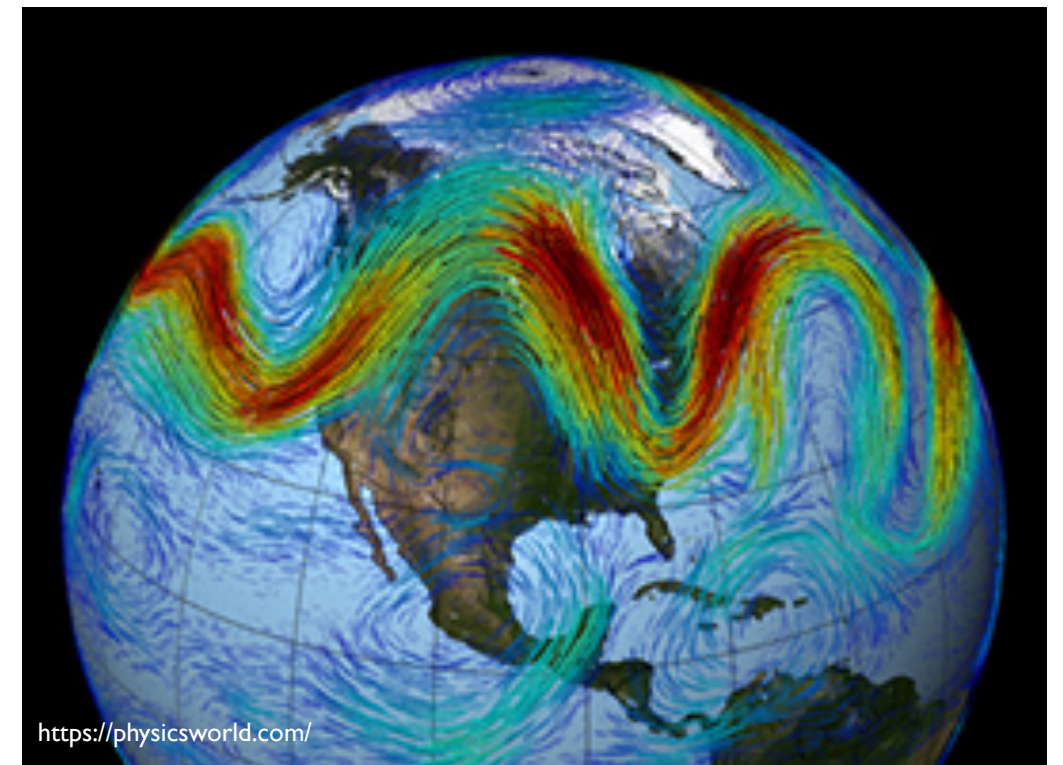
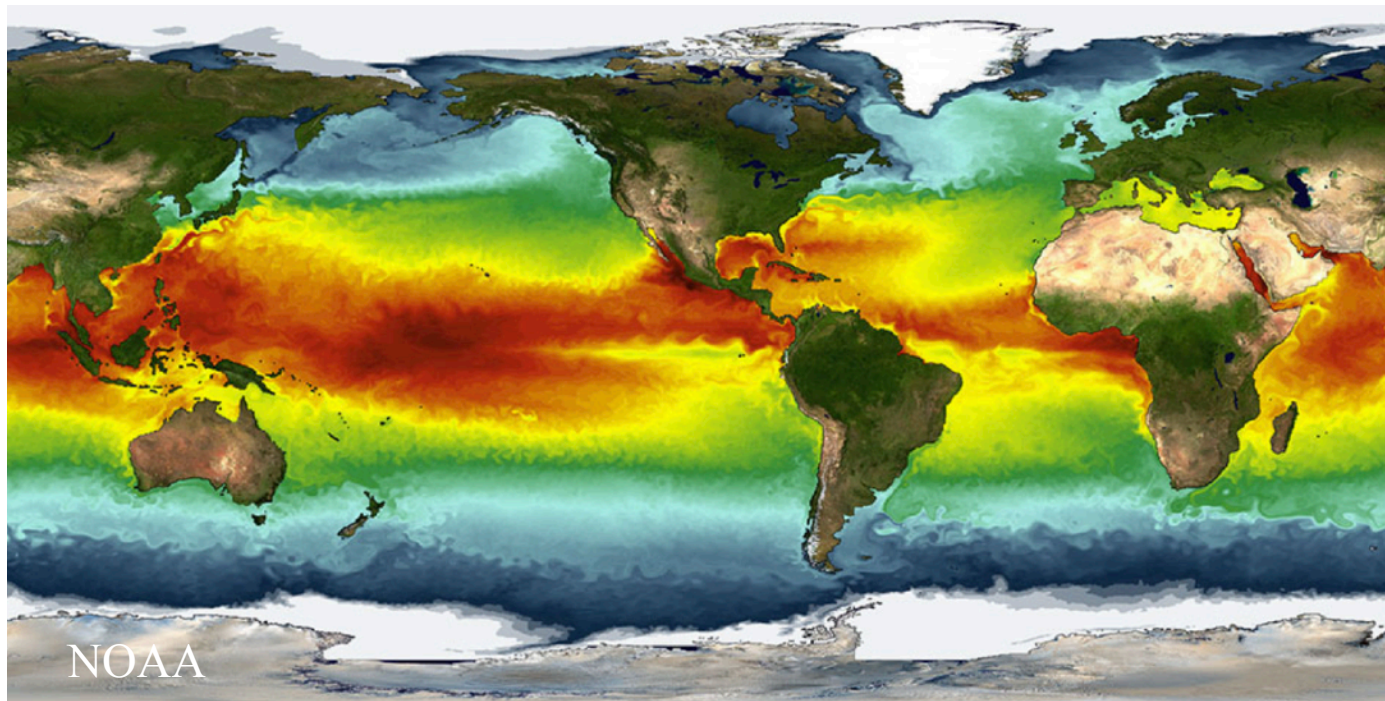
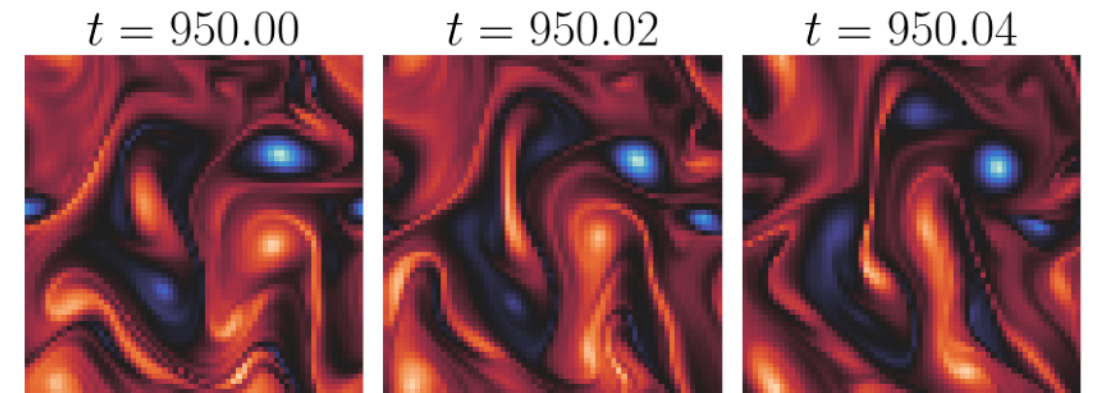
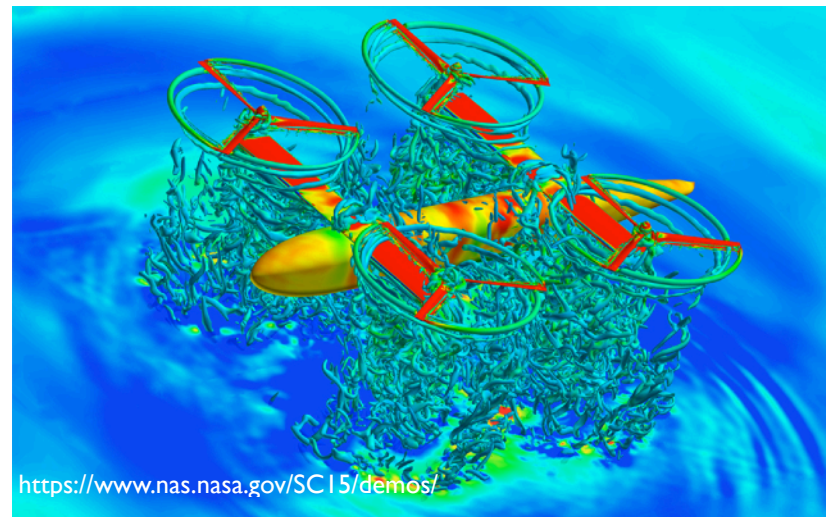
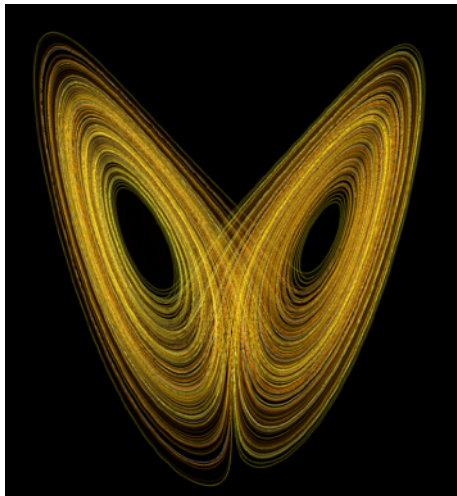
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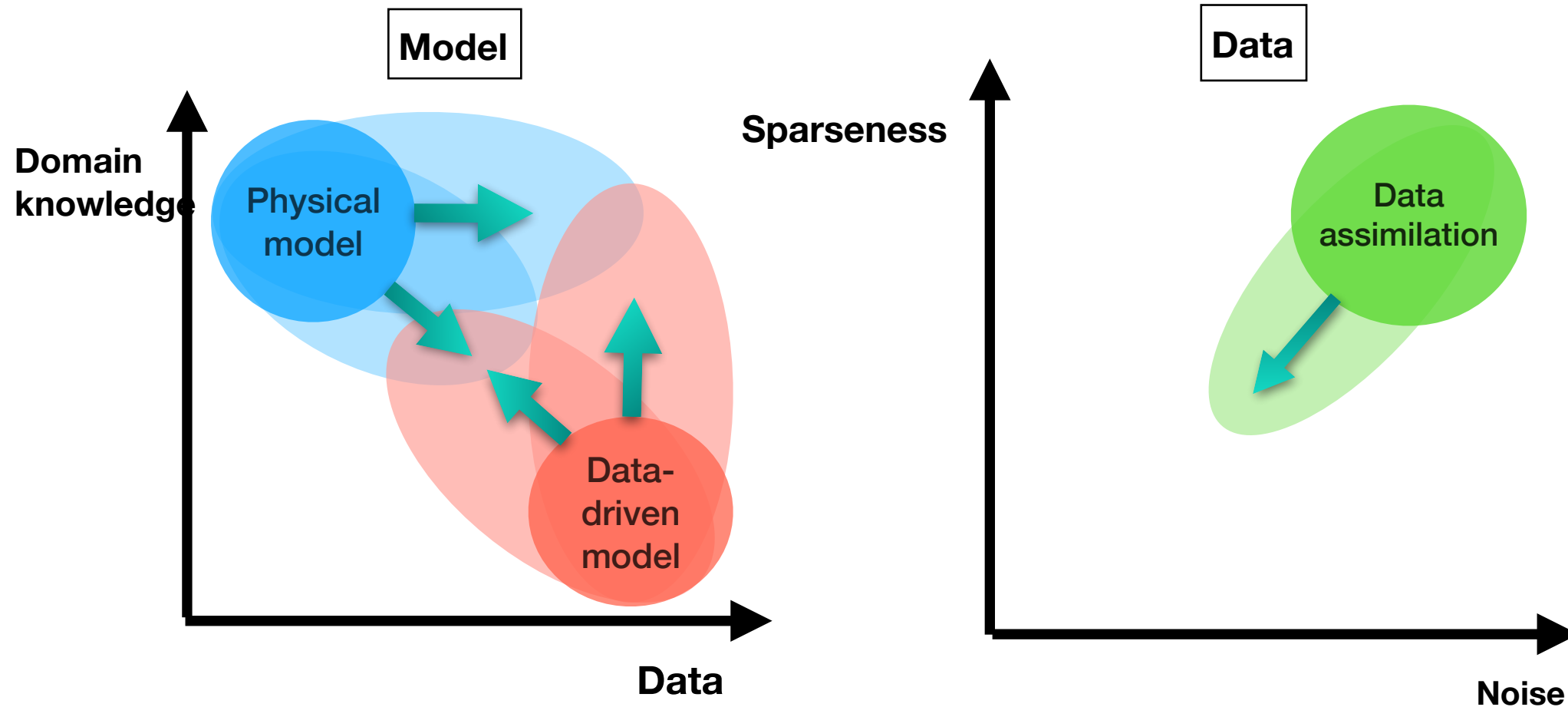
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- **Sketch review of Scientific Machine Learning (SciML) and DA**
- **Conditional Gaussian Koopman Network (CGKN):** a deep-learning framework that unifies SciML and DA
- **Lagrangian Conditional Gaussian Koopman Network (LaCGKN)** for Lagrangian Data Assimilation and Prediction



- **Complex dynamical systems:** nonlinear, chaotic, multi-scale, turbulent, intermittent, non-Gaussian; common in fluid dynamics, geophysics, neuroscience, material science...
- **Goal:**
 - Make predictions → models
 - Use observations to improve predictions → data assimilation (DA)



- Physical model:
 - Based on governing equations derived from first principles (**interpretable**)
 - May **require strong assumptions**; Usually **computationally expensive** (e.g., Numerical Weather Prediction)
- Data-driven model:
 - **Computationally efficient**; Works with governing equations unknown
 - **Lack of interpretability**; May require a large amount of data (can be sparse and noisy in reality)
- Data assimilation is especially useful when data is sparse and noisy
 - As new observations become available, DA can utilize this information to **improve real-time predictions (filtering)**
 - Combining data with existing models, DA can **recover complete data, with less uncertainty (smoothing)** → reanalysis

Data-driven models for dynamical systems (SciML)

- **Black-box surrogates:**

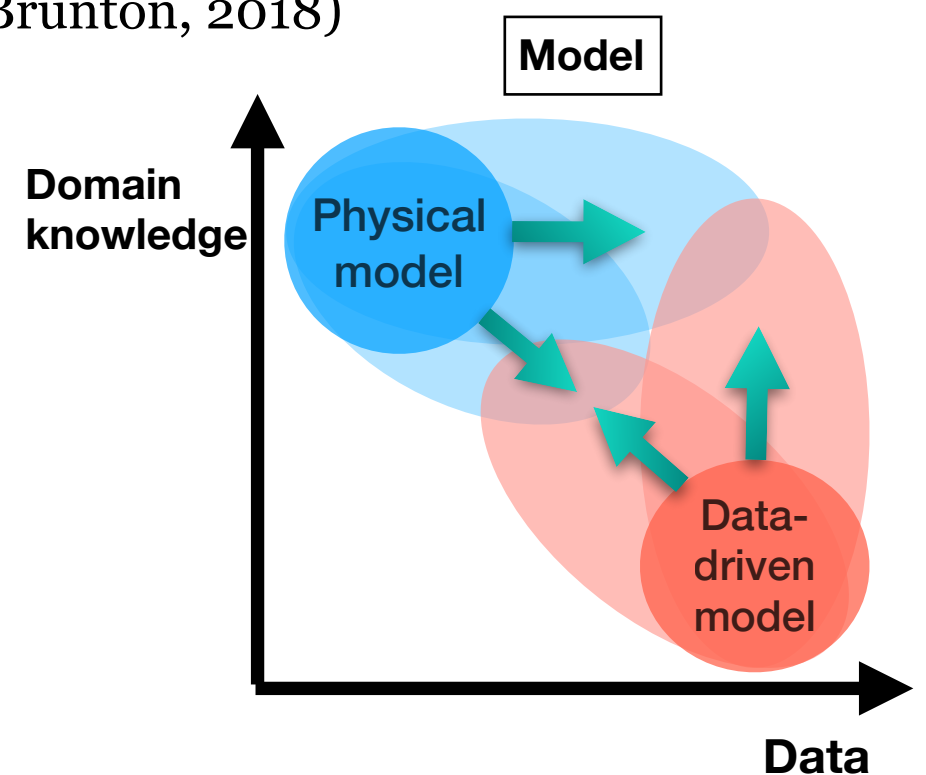
- *RNN* (Schuster & Paliwal, 1997)
- *Reservoir computing* (Gauthier et al., 2021)
- *Diffusion models, Transformers...*
- *Neural ODE* (Chen, 2018);
- *Operator learning* (Lu, 2019; Li, 2020);
- *Gaussian processes* (Chen, 2021);

- **Structure-imposing methods:**

- *Reduced-order models*: linear stochastic models
- *Dynamics identification*: Sparse dynamics approximation (Schaeffer, 2012); SINDy (Brunton, 2016)
- Koopman operator / DMD (Schmid, 2010; Lusch, Kutz, Brunton, 2018)

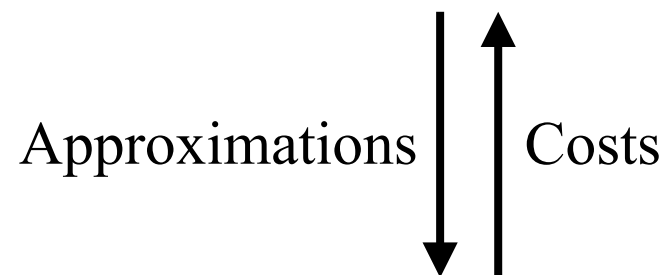
Hybrid ML with physical models:

- *Residual learning*: closure models (Rasp, Pritchard, Gentine, 2018; Levine and Stuart, 2022)
- *Physics-informed machine learning*: PINNs (Raissi, 2019)

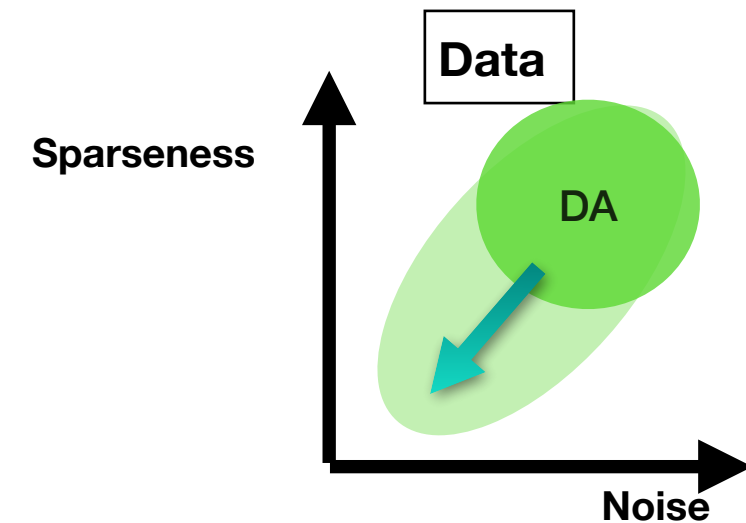


DA for using observations to improve predictions

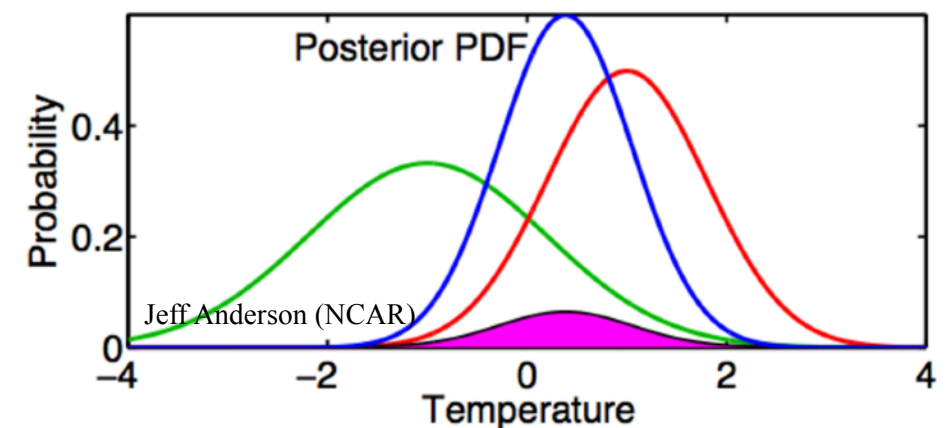
- DA is based on **Bayes' rule**. It combines model predictions and observations to get a better state estimate; especially useful when observations are **sparse** and **noisy**
- Traditional model-based DA methods:**
 - Solve the posterior using Bayes' rule **explicitly**, often relying on **linear or Gaussian assumptions** to make it parametric and computationally tractable.
 - e.g., Classical Kalman Filter (linear, Gaussian); Ensemble Kalman Filter (EnKF); Variational methods (3D/4D-Var); Particle Filter (PF)



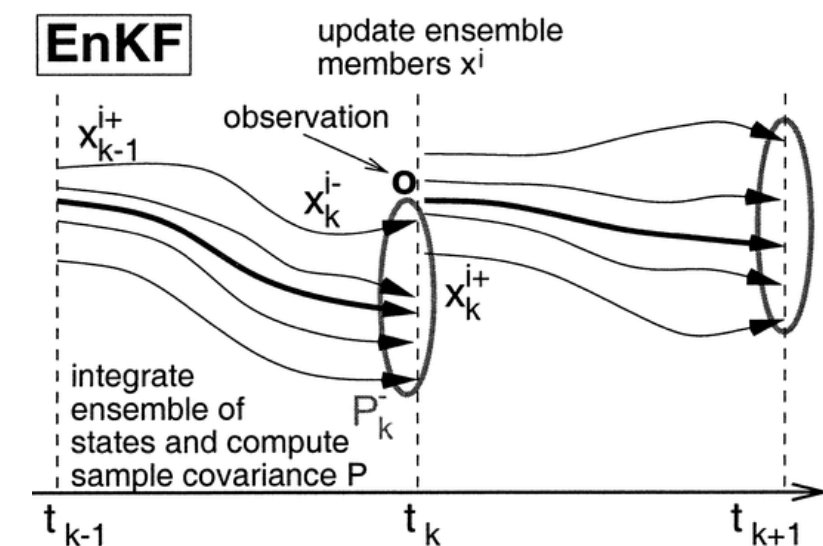
- Challenges:**
 - nonlinearity, non-Gaussianity
 - high computational costs due to high dimensionality



$$P(\mathbf{x}_{t_k} | \mathbf{Y}_{t_k}) = \frac{P(\mathbf{y}_k | \mathbf{x}) P(\mathbf{x}_{t_k} | \mathbf{Y}_{t_{k-1}})}{\text{Normalization}}$$



Bayes rule (1D Gaussian case).



[Reichle, R. H., 2002]

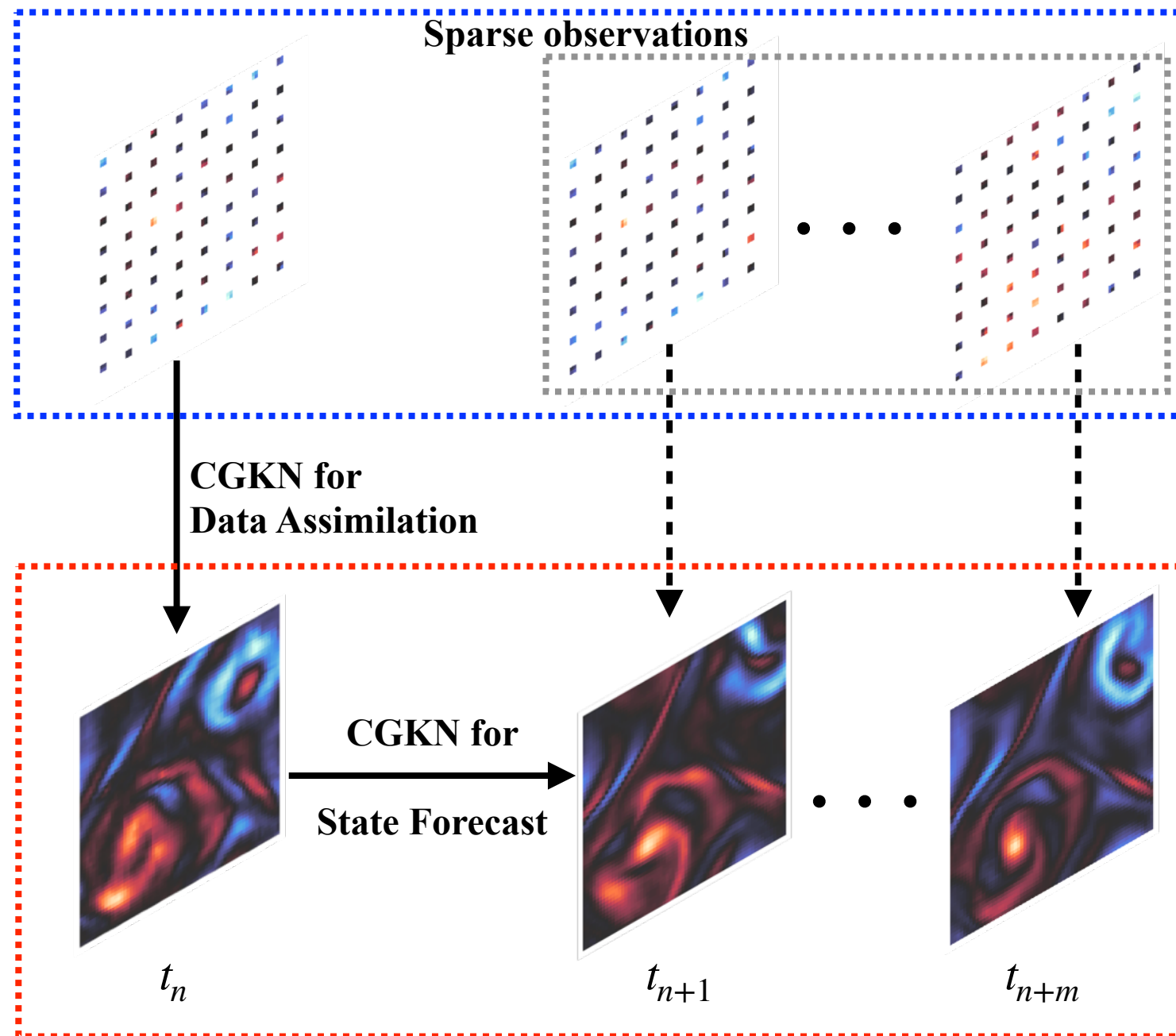
Combining SciML with DA

- **Data-enhanced/driven DA methods** (ML for DA)
 - ML models as **cheap surrogate models** to generate ensembles; **model error correction**; **learning parameters** of an (parameterized) analysis map;
 - Pure **data-driven DA**, e.g., generative DA based on conditional sampling, transport maps
 - Hybrid approaches that **preserve analytically tractable nonlinear structures**, e.g., CGNSDE (Chen et al., 2024), CGKN (Chen et al. 2025)
- **DA for ML**
 - DA analysis as **better training data**
 - **Online correction of ML model predictions**
 - **Derivative-free optimization** (EKI; Iglesias, 2013)

[Review: Cheng et al., 2023; Bach et al., 2025]

- **Reviewing Scientific Machine Learning (SciML) and DA**
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Conditional Gaussian Koopman Network (CGKN)



- A **deep learning digital twins framework** that unifies SciML and DA, to learn surrogate models that performs **efficient DA** and **prediction (with UQ)** simultaneously for **nonlinear partially observed** dynamical systems.

[CGKN: A deep learning framework for modeling complex dynamical systems and efficient data assimilation. Chen et al. 2025]

[Modeling partially observed nonlinear dynamical systems and efficient data assimilation via discrete-time conditional Gaussian Koopman network, Chen et al. , 2025]

Motivation from DA (CG filter):

$\mathbf{u}_1$: observed states
 $\mathbf{v}$: unobserved states

Conditional Gaussian Nonlinear System

$$\frac{d\mathbf{u}_1}{dt} = \mathbf{f}_1(\mathbf{u}_1) + \mathbf{g}_1(\mathbf{u}_1)\mathbf{v} + \sigma_1 \dot{\mathbf{W}}_1$$

$$\frac{d\mathbf{v}}{dt} = \mathbf{f}_2(\mathbf{u}_1) + \mathbf{g}_2(\mathbf{u}_1)\mathbf{v} + \sigma_2 \dot{\mathbf{W}}_2$$

CGNS gallery

- The noisy Lorenz model
- Boussinesq equation
- Rotating shallow water equations
- The predator-prey model
- ...

[Chen and Majda, 2018]

CG filter

- The posterior distribution is Gaussian given the past observations up to time t .

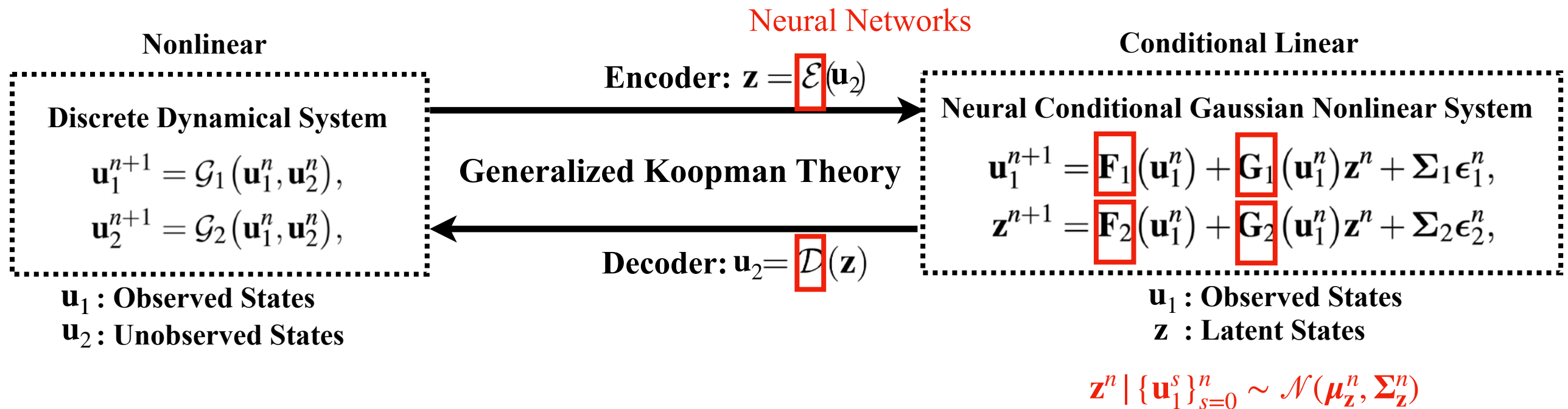
$$\mathbf{v}(t) | \mathbf{u}_1(s), s \leq t \sim \mathcal{N}(\boldsymbol{\mu}_v(t), \mathbf{R}_v(t))$$

- The posterior mean and covariance can be solved via **analytic formulae**.

$$\frac{d\boldsymbol{\mu}_v}{dt} = (\mathbf{f}_2 + \mathbf{g}_2\boldsymbol{\mu}_v) + (\mathbf{R}_v\mathbf{g}_1^T)(\sigma_1\sigma_1^T)^{-1} \left(\frac{d\mathbf{u}_1}{dt} - (\mathbf{f}_1 + \mathbf{g}_1\boldsymbol{\mu}_v) \right)$$

$$\frac{d\mathbf{R}_v}{dt} = \mathbf{g}_2\mathbf{R}_v + \mathbf{R}_v\mathbf{g}_2^T + \sigma_2\sigma_2^T - \mathbf{R}_v\mathbf{g}_1^T(\sigma_1\sigma_1^T)^{-1}(\mathbf{g}_1\mathbf{R}_v)$$

Motivation from Koopman theory:



- The latent representation $\mathbf{z}$ is conditional linear given $\mathbf{u}_1$ being observed, which leads to a neural conditional Gaussian nonlinear system (CGNS) that has [analytic DA formulae](#).
- Instead of directly applying Koopman theory, CGKN only seeks for embeddings of the unobserved states $\mathbf{u}_2$.

CGKN (Discrete-time)

Nonlinear

Neural Networks

Conditional Linear

Discrete Dynamical System

$$\begin{aligned} \mathbf{u}_1^{n+1} &= \mathcal{G}_1(\mathbf{u}_1^n, \mathbf{u}_2^n), \\ \mathbf{u}_2^{n+1} &= \mathcal{G}_2(\mathbf{u}_1^n, \mathbf{u}_2^n), \end{aligned}$$

$\mathbf{u}_1$: Observed States
 $\mathbf{u}_2$: Unobserved States

Encoder: $\mathbf{z} = \mathcal{E}(\mathbf{u}_1)$

Generalized Koopman Theory

Decoder: $\mathbf{u}_1 = \mathcal{D}(\mathbf{z})$

Neural Conditional Gaussian Nonlinear System

$$\begin{aligned} \mathbf{u}_1^{n+1} &= \mathbf{F}_1(\mathbf{u}_1^n) + \mathbf{G}_1(\mathbf{u}_1^n)\mathbf{z}^n + \Sigma_1\epsilon_1^n, \\ \mathbf{z}^{n+1} &= \mathbf{F}_2(\mathbf{u}_1^n) + \mathbf{G}_2(\mathbf{u}_1^n)\mathbf{z}^n + \Sigma_2\epsilon_2^n, \end{aligned}$$

$\mathbf{u}_1$: Observed States
 $\mathbf{z}$: Latent States

$$\mathbf{z}^n | \{\mathbf{u}_1^s\}_{s=0}^n \sim \mathcal{N}(\mu_z^n, \Sigma_z^n)$$

CG filter

(Efficient Analytical Formulae)

$$\mu_z^n = \mathbf{F}_2 + \mathbf{G}_2\mu_z^{n-1} + \mathbf{K}^{n-1}(\mathbf{u}_1^n - \mathbf{F}_1 - \mathbf{G}_1\mu_z^{n-1})$$

Data Assimilation

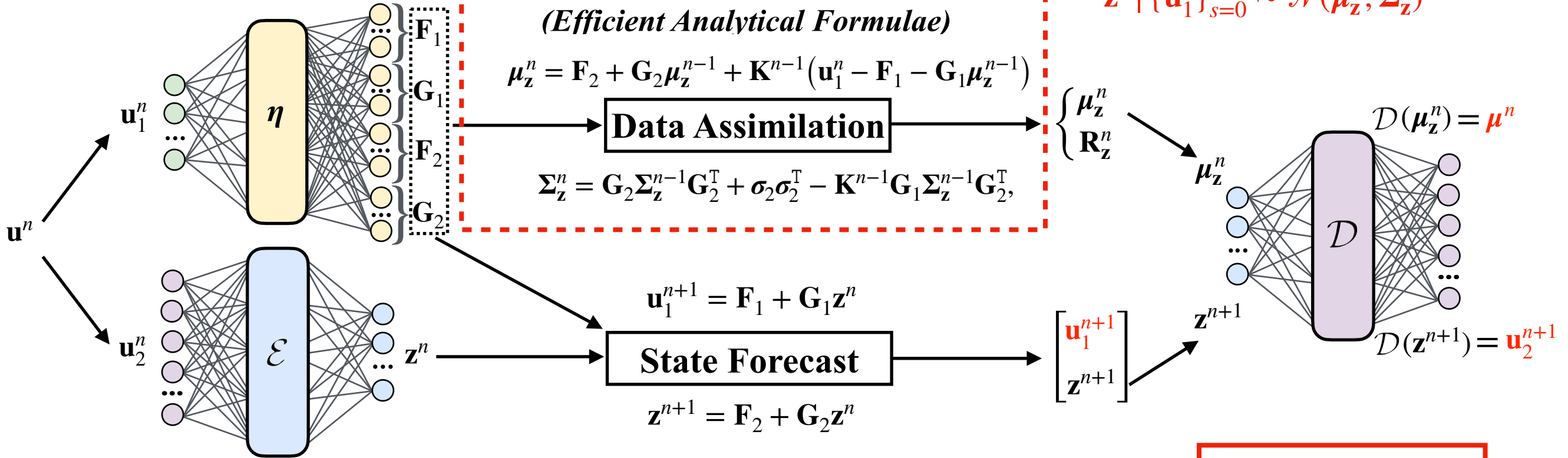
$$\Sigma_z^n = \mathbf{G}_2\Sigma_z^{n-1}\mathbf{G}_2^T + \sigma_2\sigma_2^T - \mathbf{K}^{n-1}\mathbf{G}_1\Sigma_z^{n-1}\mathbf{G}_2^T,$$

$$\begin{Bmatrix} \mu_z^n \\ \mathbf{R}_z^n \end{Bmatrix}$$

$$\mu_z^n$$

$$\mathcal{D}(\mu_z^n) = \mu^n$$

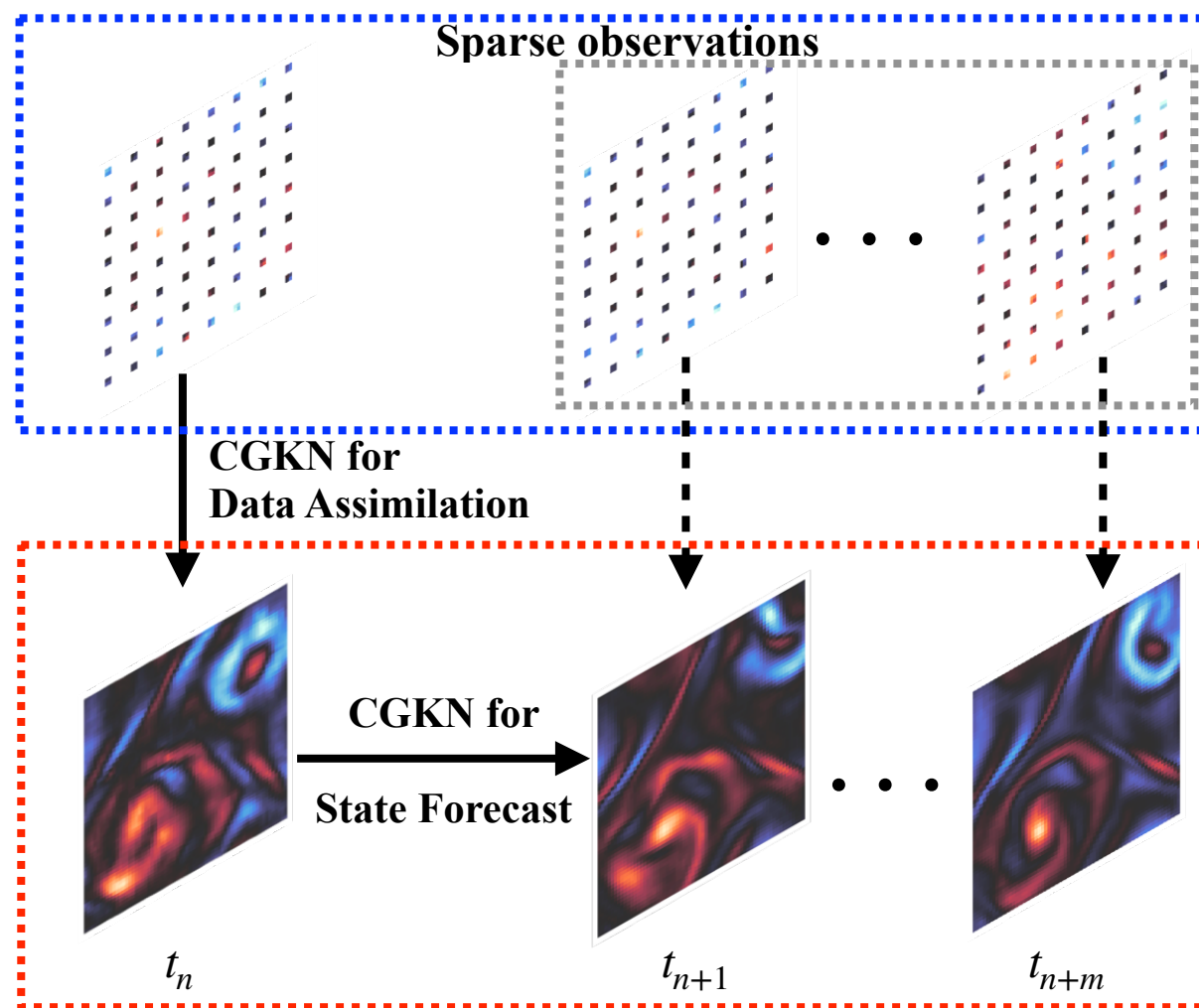
$$\mathcal{D}(\mathbf{z}^{n+1}) = \mathbf{u}_2^{n+1}$$



$$\underbrace{L(\theta_{\mathcal{E}}, \theta_{\mathcal{D}}, \theta_{\eta})}_{\text{Total loss}} := \underbrace{\lambda_{\text{AE}} L_{\text{AE}}(\theta_{\mathcal{E}}, \theta_{\mathcal{D}})}_{\text{Auto-encoder loss}} + \underbrace{\lambda_{\mathbf{u}} L_{\mathbf{u}}(\theta_{\mathcal{E}}, \theta_{\mathcal{D}}, \theta_{\eta})}_{\text{Forecast loss of physical variables}} + \underbrace{\lambda_{\mathbf{z}} L_{\mathbf{z}}(\theta_{\mathcal{E}}, \theta_{\eta})}_{\text{Forecast loss of latent variables}} + \underbrace{\lambda_{\text{DA}} L_{\text{DA}}(\theta_{\mathcal{D}}, \theta_{\eta})}_{\text{Data assimilation loss}}$$

- State forecast and DA are performed in the latent space (with **reduced dimension**)
- DA formulae is part of the model structure as **inductive bias**, and learning is **constrained by DA loss**.

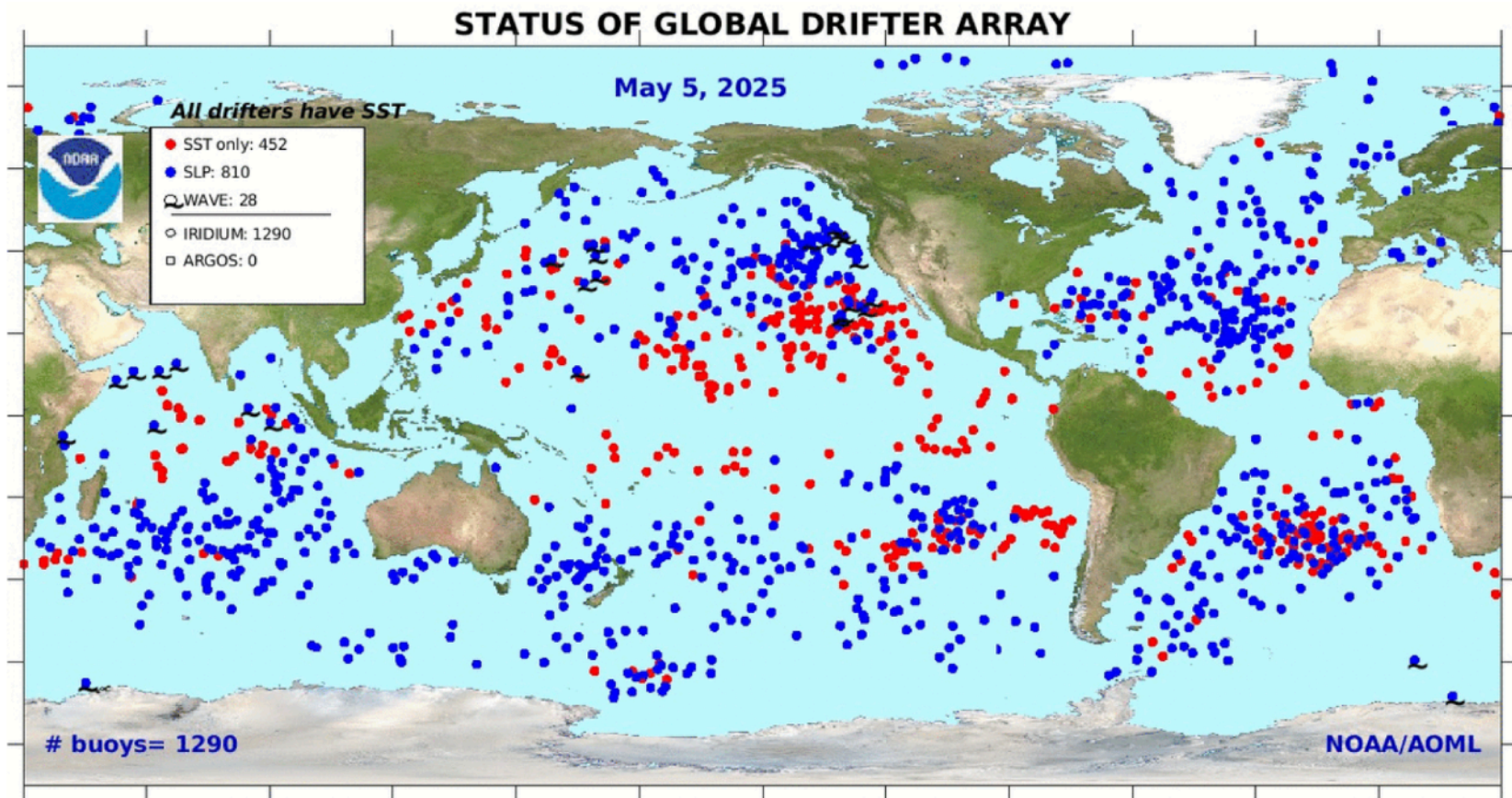
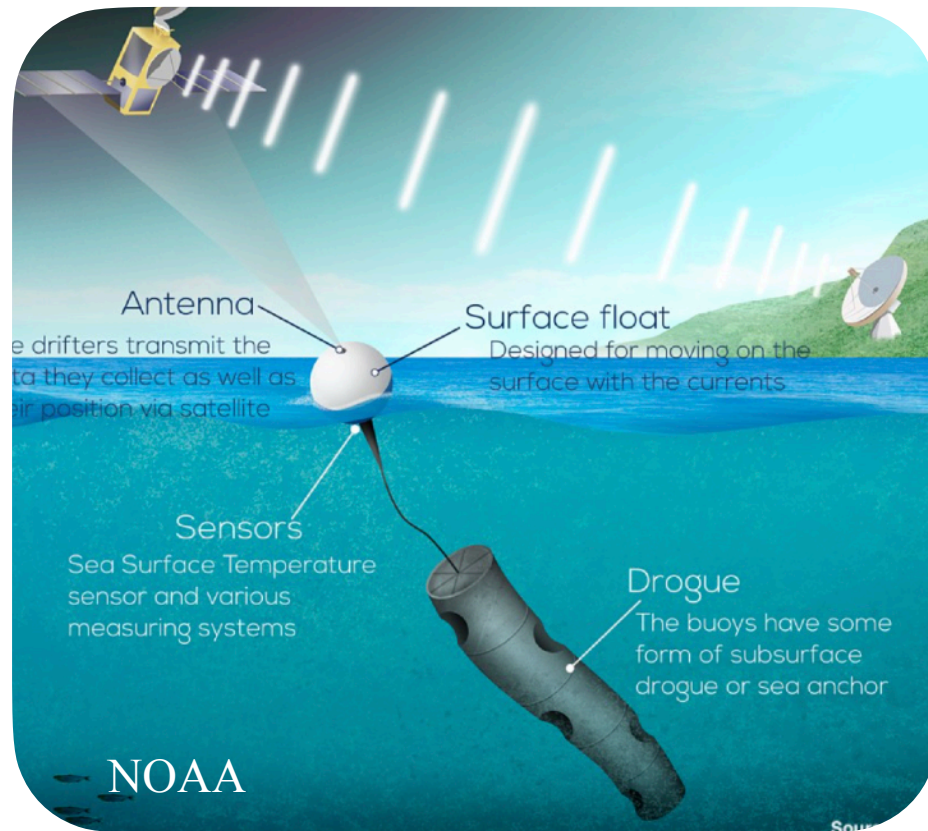
CGKN (Discrete-time)



- Make predictions with only **partially and noisy observed initial conditions**, and **improve predictions as new observations come (DA)**
- *Analytic DA* formulae of CG filter (model-based DA):
 - Ensures **accuracy** and **efficiency** of DA (avoid using ensembles)
 - Introduces **inductive bias** to model structure, and constrains the learning
 - Also facilitates the downstream tasks like **uncertainty quantification (UQ)**
- Bridges model-based DA and data-driven DA, leading to a **unified framework for SciML and DA**.

- **Reviewing Scientific Machine Learning (SciML) and DA**
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Lagrangian data assimilation



- Lagrangian DA uses **Lagrangian** observations to recover the underlying **Eulerian** flow.
 - e.g., inferring deep ocean states with surface observations; inferring upper atmosphere with ground-surface observations
- Lagrangian DA has **increasing practical importance** due to a growing amount of Lagrangian observations in operational oceanic and atmospheric observing systems.

Lagrangian DA: a canonical example

- Consider Lagrangian observations of I passive tracer positions driven by an hidden Eulerian flow:

$$\text{Tracer: } \mathbf{x}_i^{n+1} = \mathcal{T}_h(\mathbf{x}_i^n, \mathbf{v}^n) + \Sigma_{\mathbf{x}_i} \Delta \mathbf{W}_i^n, \quad i = 1, \dots, I,$$

$$\text{Flow: } \mathbf{v}^{n+1} = \mathcal{F}_h(\mathbf{v}^n) + \Sigma_{\mathbf{v}} \Delta \mathbf{W}_{\mathbf{v}}^n,$$

- The Lagrangian filtering problem aims to find the posterior of the unobserved flow given past tracer observations:

$$p(\mathbf{v}^n | \{\mathbf{x}_i^s\}_{i=1, s=0}^{I, n})$$

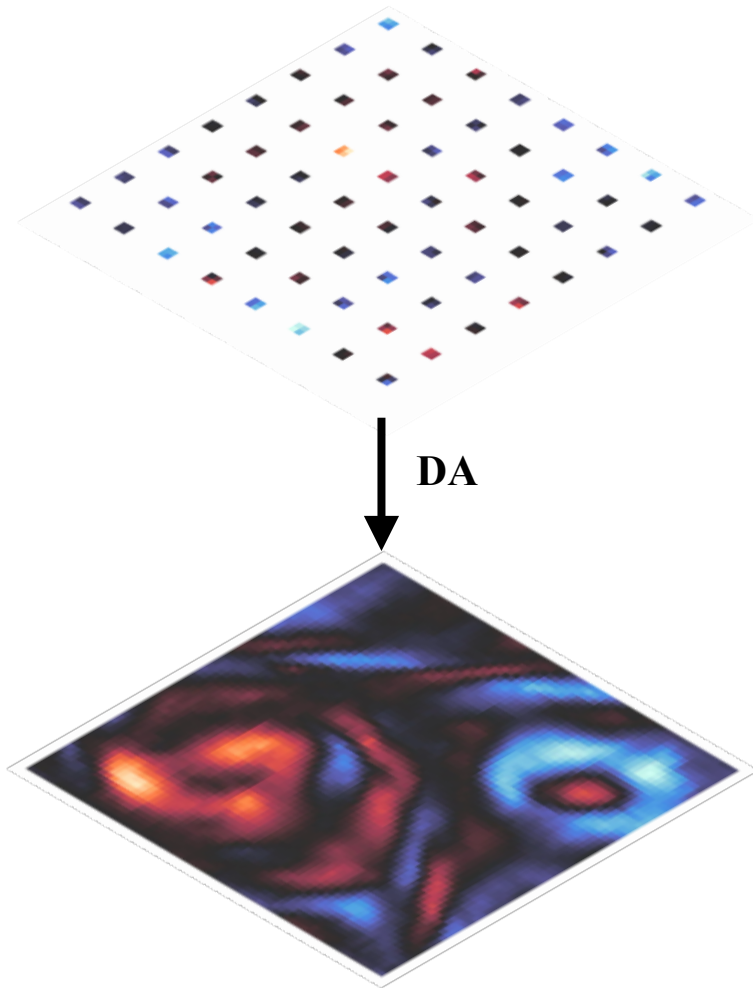
- The tracer operator $\mathcal{T}_h$ involves an interpolation or integral operator that evaluates flow $\mathbf{v}$ at a local tracer position $\mathbf{x}_i$, which is typically **nonlinear**.
- The **observation process** of Lagrangian DA is **intrinsically nonlinear**, making it more **challenging** than Eulerian DA.

Lagrangian DA: approaches

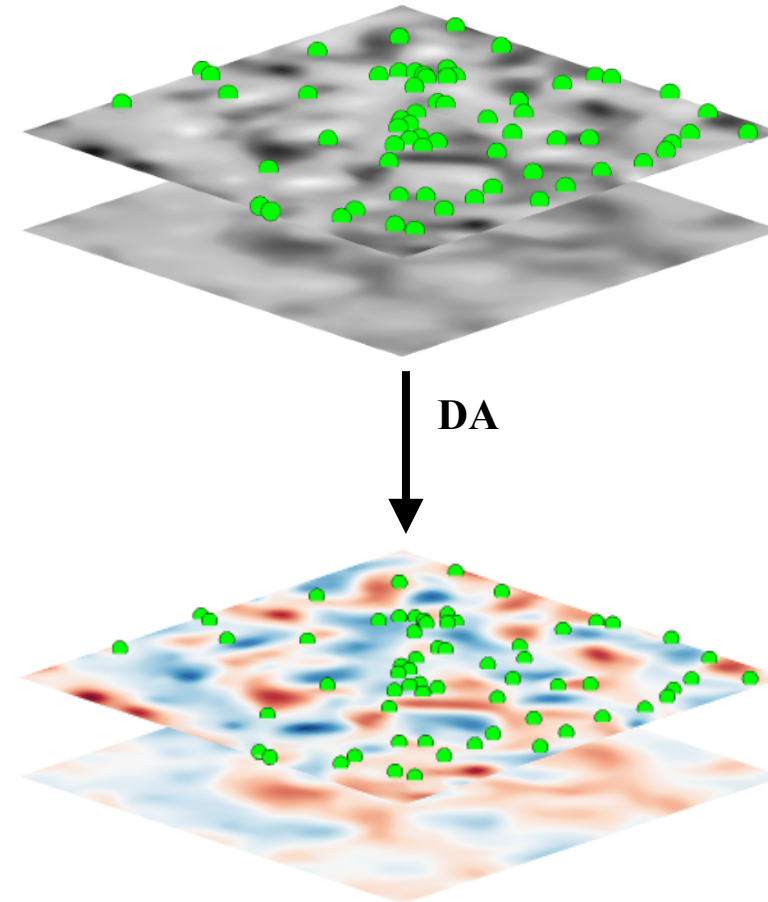
- **Traditional deductive (model-based) methods**
 - **Kalman-based methods**: extended Kalman filter (EKF; Ide et al. 2002); optimal interpolation (OI; Molcard et al. 2003); local ensemble transform Kalman filter (LETKF; Sun & Penny, 2019)
 - **Partical filters (PF)**, Markov chain Monte Carlo (**MCMC**) **smoothers** (Apte et al., 2008)
 - **Hybrid approaches**, e.g., combining PF for nonlinear tracer dynamics with EnKF for near-Gaussian flow estimation
- **ML inductive (data-driven) methods**
 - **Combining neural operators with generative models** (Asefi et al., 2025)
 - **Multimodal contrastive learning** (Baptista et al., 2025)
- **ML for Lagrangian DA remains relatively limited.**
- ML is particularly well suited to **addressing the strong nonlinearity** inherent in Lagrangian DA, holding **substantial potential to improve accuracy and efficiency.**

CGKN for Lagrangian DA: Challenges

CGKN for N-S flow with
direct flow observations

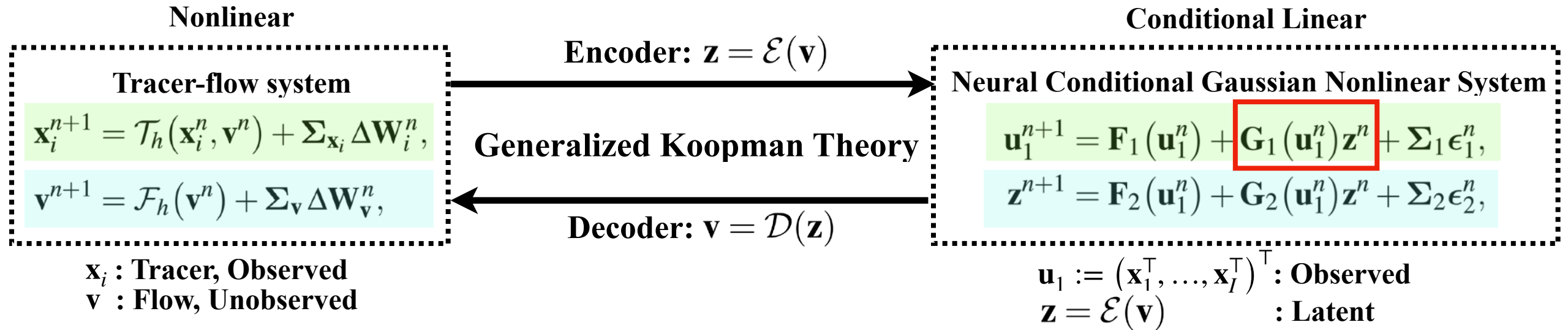


LaCGKN for two-layer QG flow with
indirect Lagrangian tracer observations



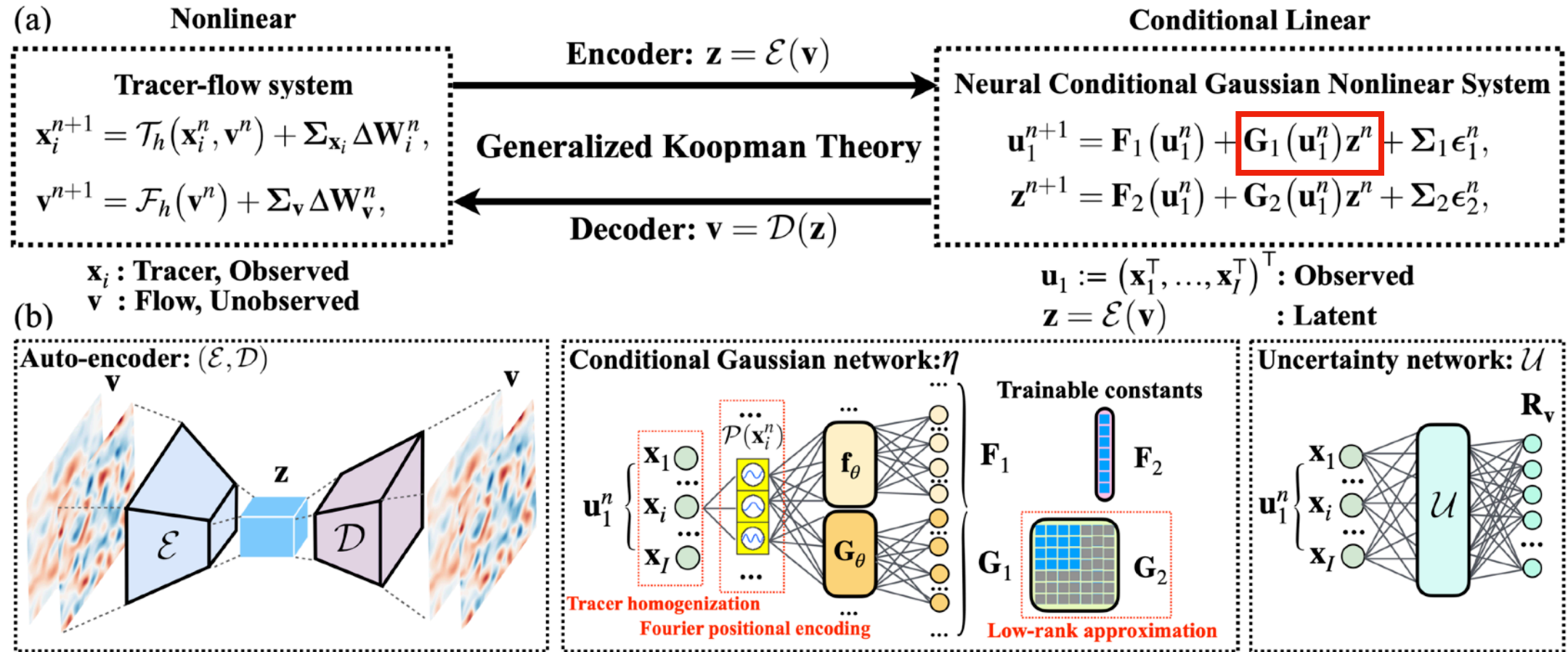
- Another challenge: compared to the earlier application of CGKN to N-S equations with *direct* observations of the flow, Lagrangian DA uses **sparse** and *indirect* Lagrangian observations.

CGKN for Lagrangian DA: Challenges



- Lagrangian DA is thus a more **challenging yet practical** problem **for CGKN**:
 - not only the **nonlinear flow dynamics of $\mathbf{v}$** needs to be well approximated by a **latent linear dynamics of $\mathbf{z}$** (as in standard Koopman theory)
 - but also the **nonlinear tracer dynamics** should be well approximated by the **neural tracer dynamics driven by latent flow**. $\rightarrow \mathbf{G}_1(\mathbf{u}_1^n) \mathbf{z}^n$
- **The latter is crucial to data assimilation**, as it captures the information propagation from observed states to unobserved states. This **imposes additional requirements and regularizations** for **latent embedding $\mathbf{z}$** .

LaCGKN structure design



1. **Homogenization over tracers.** Lagrangian tracers can often be assumed to be homogeneous. We therefore construct $\mathbf{F}_1$ and $\mathbf{G}_1$ by applying the same neural networks to each tracer position independently:

$$\mathbf{F}_1(\mathbf{u}_1^n) := (\mathbf{f}_\theta(\mathbf{x}_1^n)^\top, \dots, \mathbf{f}_\theta(\mathbf{x}_I^n)^\top)^\top,$$

$$\mathbf{G}_1(\mathbf{u}_1^n) := (\mathbf{G}_\theta(\mathbf{x}_1^n)^\top, \dots, \mathbf{G}_\theta(\mathbf{x}_I^n)^\top)^\top,$$

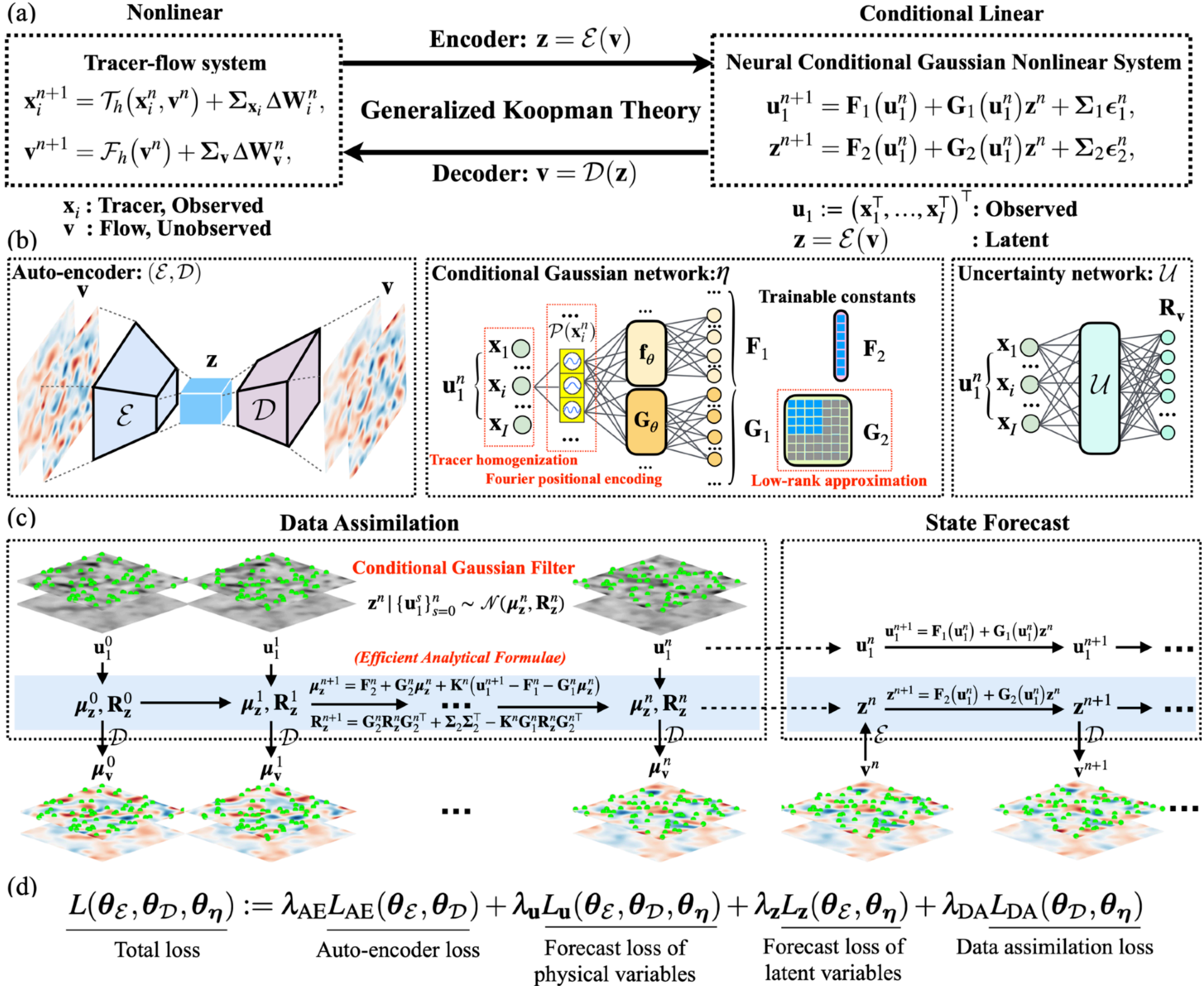
2. **Fourier positional encoding.** To accurately reconstruct the local value of the flow field at moving tracer locations, $\mathbf{G}_1$ must learn a rich nonlinear dependence on position $\mathbf{x}$. We therefore adopt the Fourier positional encoding:

$$\mathcal{P}(\mathbf{x}_i^n) = \left[x_{i,1}^n, \dots, x_{i,d}^n, \left\{ \sin(2^k \pi x_{i,j}^n), \cos(2^k \pi x_{i,j}^n) \right\}_{j=1, k=0}^{d, K-1} \right]$$

3. **Low-rank approximation of $\mathbf{G}_2$.** To control the computational complexity while the latent dimension scales up, a SVD-inspired low-rank approximation of $\mathbf{G}_2$ is adopted:

$$\mathbf{G}_2 = \mathbf{U} \text{diag}(\mathbf{s}) \mathbf{V}^\top + \text{diag}(\delta)$$

LaCGKN overview



Numerical tests

- **Tested case:** A two-layer quasi-geostrophic (QG) flow with passive tracer position observations (Tracers are advected by the upper-layer flow).

$$\text{Tracer: } \left\{ \frac{d\mathbf{x}_i}{dt} = \mathbf{v}(\mathbf{x}_i, t) + \Sigma_{\mathbf{x}_i} \dot{\mathbf{W}}_i, \quad i = 1, \dots, I \right.$$

$$\text{Flow: } \left\{ \begin{aligned} \frac{\partial q_1}{\partial t} + J(\psi_1, q_1) + \beta \frac{\partial \psi_1}{\partial x} + U_1 \frac{\partial}{\partial x} \nabla^2 \psi_1 + \frac{k_d^2}{2} \left(U_1 \frac{\partial \psi_2}{\partial x} - U_2 \frac{\partial \psi_1}{\partial x} \right) &= -\nu \Delta^s q_1, \\ \frac{\partial q_2}{\partial t} + J(\psi_2, q_2) + \beta \frac{\partial \psi_2}{\partial x} + U_2 \frac{\partial}{\partial x} \nabla^2 \psi_2 + \frac{k_d^2}{2} \left(U_2 \frac{\partial \psi_1}{\partial x} - U_1 \frac{\partial \psi_2}{\partial x} \right) &= - \left(U_2 \frac{\partial h}{\partial x} + \kappa \nabla^2 \psi_2 \right) - \nu \Delta^s q_2, \\ q_1 = \nabla^2 \psi_1 + \frac{k_d^2}{2} (\psi_2 - \psi_1), \quad q_2 = \nabla^2 \psi_2 + \frac{k_d^2}{2} (\psi_1 - \psi_2) + h, \quad \mathbf{v}_\ell &= \left(\frac{\partial \psi_\ell}{\partial y}, -\frac{\partial \psi_\ell}{\partial x} \right)^\top \end{aligned} \right.$$

Quantity of interest (to be recovered): stream function $\{\psi_1, \psi_2\}$

- **Compared benchmarks:**

- LaCGKN

- DNN for tracer + CNN for flow.

(Predication)

- Persistence

- Ensemble Kalman Filter (EnKF)

(Data assimilation)

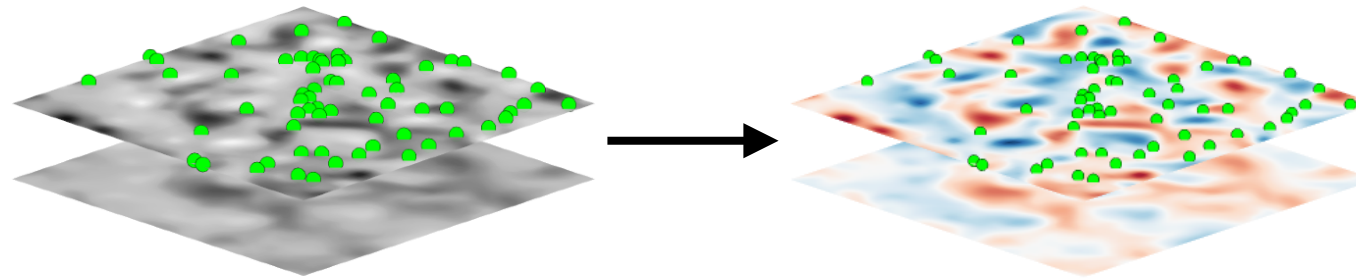
- Optimal interpolation (OI)

- Climatology.

Two-layer QG flow with surface tracer observations

Experimental Settings

- Parameters: $k_d = 10$, $\beta = 22$, $U = 1$, $\kappa = 9$, $\nu = 10^{-12}$, and $h(x, y) = 40\cos x + 80\cos(2y)$
- Data generated on 128×128 pseudo-spectral grid over $\Omega = [0, 2\pi)^2$, $N_t = 2 \times 10^6$ time steps, $\Delta t = 2 \times 10^{-3}$
- Data are then sub-sampled to 64×64 grids, $\Delta t_{\text{obs}} = 4 \times 10^{-2}$
- Training/validation/test data: 80,000/10,000/10,000 steps
- Fourier positional encoding with $K = 6$ frequencies. Low-rank approximation of $\mathbf{G}_2$ with effective rank $r = 64$
- CGKN hyperparameters: $N_s = 1$, $N_l = 100$, $N_b = 20$, $\lambda_{\text{AE}} = \lambda_{\mathbf{u}} = \lambda_{\mathbf{v}} = \lambda_{\text{DA}} = 1$.
- EnKF ensemble size=40; Both EnKF and OI use the perfect QG model for flow forecast
- Observation: positions of **64** tracers with measurement noise $\sim \mathcal{N}(0, 0.01^2)$;
unobserved states: **$64 \times 64 \times 2$** (two-layer flow)



- Encoder $\mathbb{R}^{64 \times 64 \times 2} \mapsto \mathbb{R}^{16 \times 16 \times 2}$ / $\mathbb{R}^{64 \times 64 \times 2} \mapsto \mathbb{R}^{32 \times 32 \times 2}$, decoder mirrors encoder
(LaCGKN) (LaCGKN₃₂)

Summary of numerical results

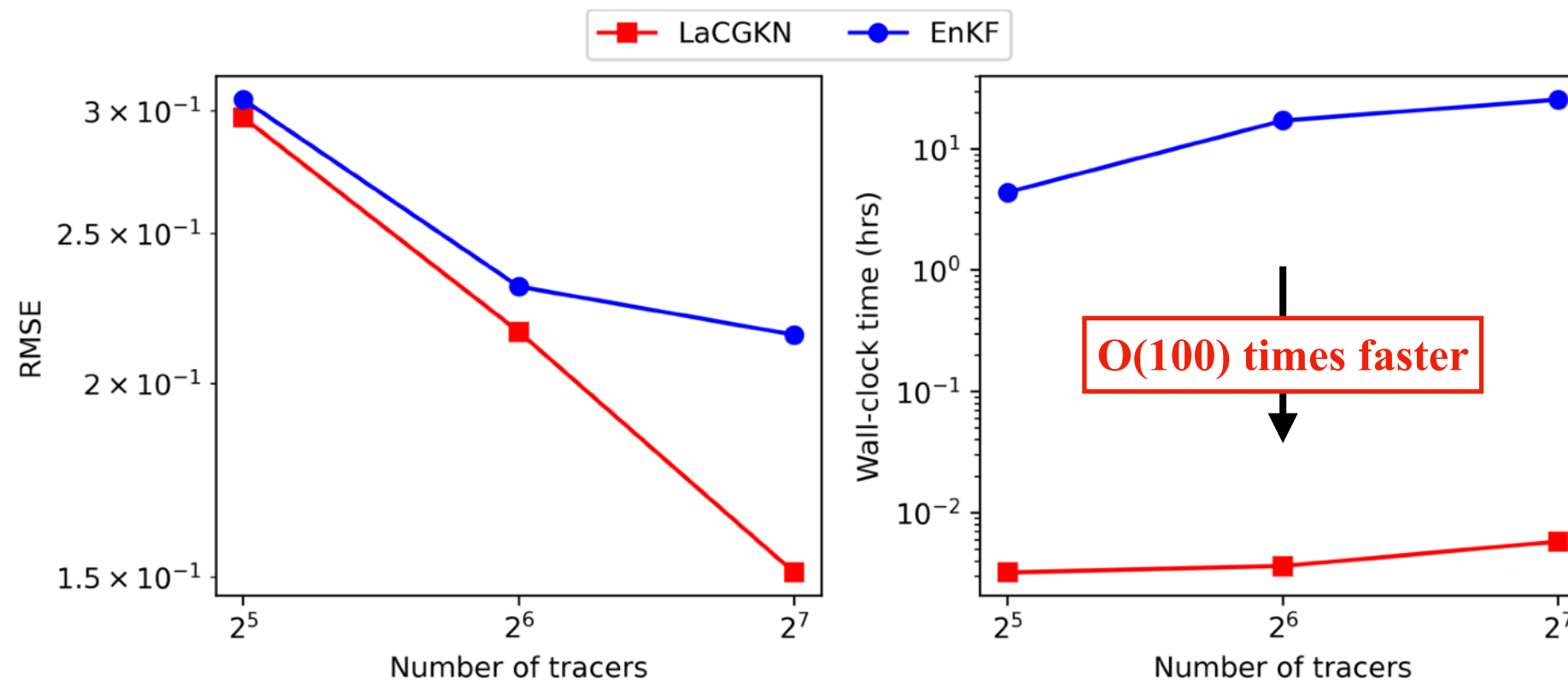
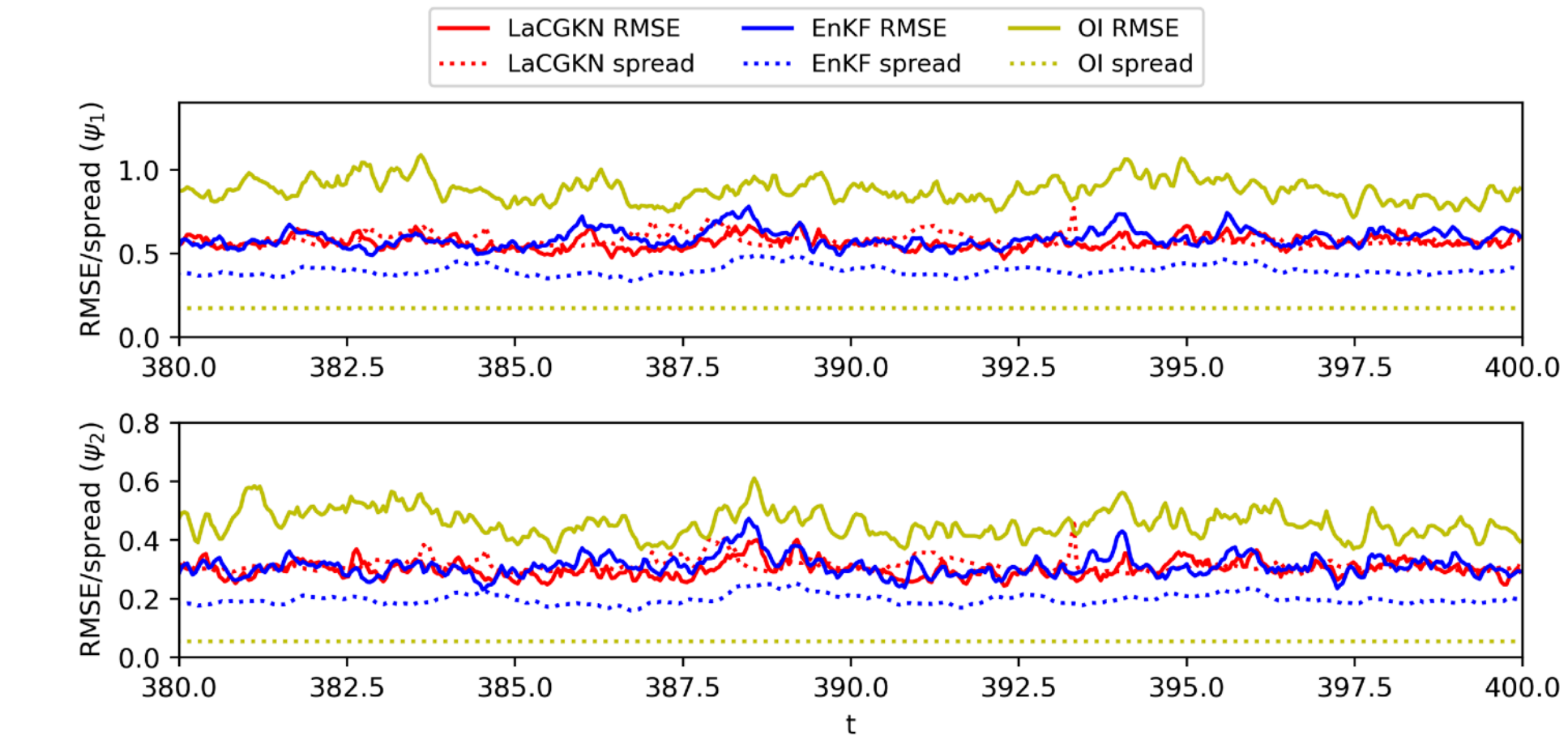
Table 2: Relative RMSEs of state forecast (one-step prediction)

Method	Tracer	Upper Layer	Lower Layer	Two Layers
LaCGKN	0.099	0.125	0.079	0.104
LaCGKN ₃₂	0.094	0.042	0.032	0.037
DNN+CNN	0.064	0.071	0.069	0.070
Persistence	0.136	0.294	0.177	0.243

Table 4: Relative RMSEs of data assimilation posterior estimates.

Method	Upper Layer	Lower Layer	Two Layers
LaCGKN	0.579	0.310	0.464
EnKF	0.599	0.321	0.481
OI	0.890	0.467	0.710
Climatology	0.870	0.414	0.681

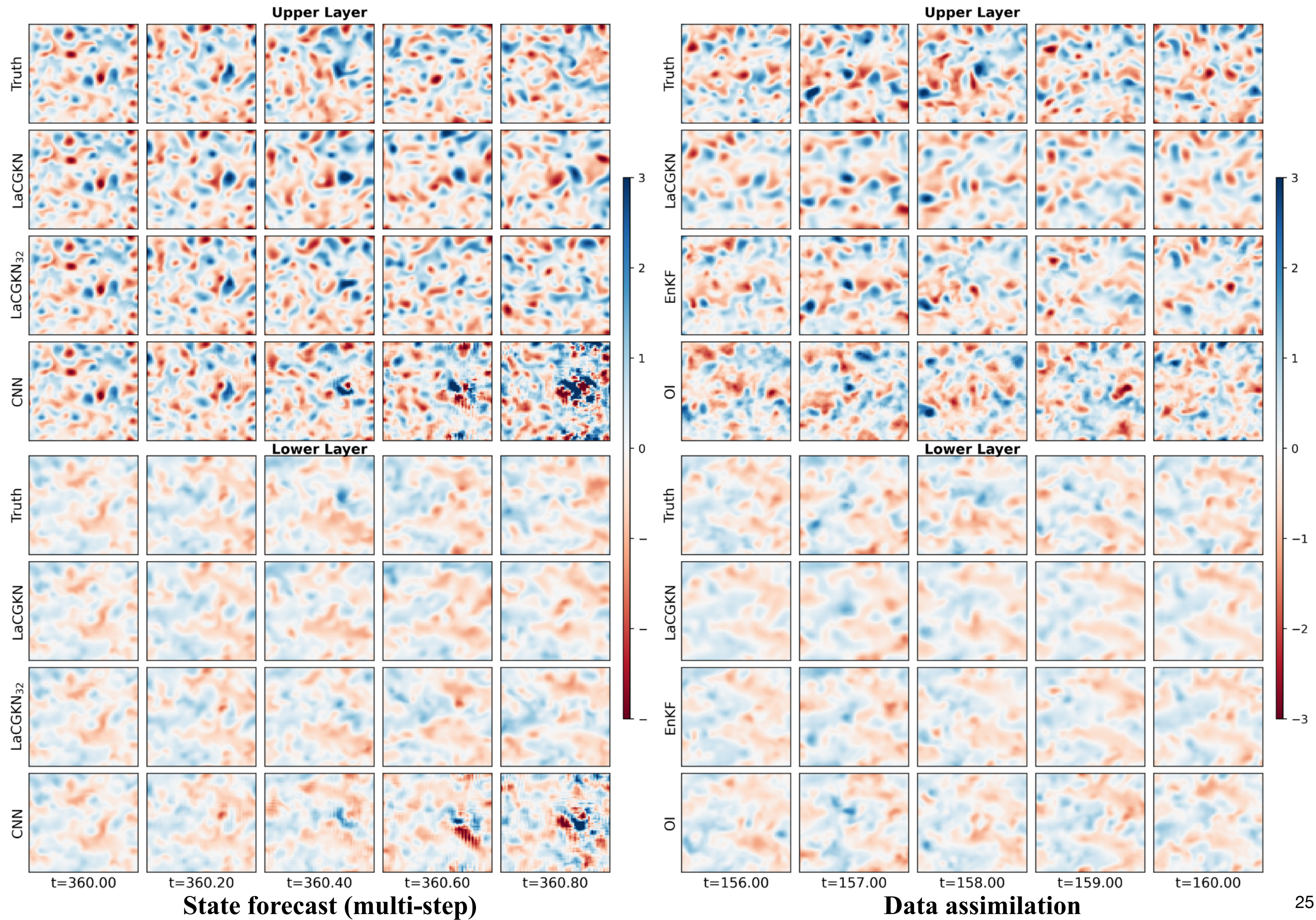
DA Performance (with UQ): Accuracy and Efficiency



Accuracy

Efficiency

Two-layer QG flow with surface tracer observations



Summary

- **CGKN** is a **unified deep learning framework** for efficient state forecast and **DA**.
- **LaCGKN** applies to Lagrangian DA with **indirect partial observations** that are **intrinsically nonlinearly coupled** with hidden states.
- **Several innovative design** to address the nonlinear tracer–flow coupling and high dimensionality, including (1) **tracer homogenization**, (2) **Fourier positional encoding**, (3) **low-rank parameterization**.
- **An application** to the two-layer QG flow with surface tracer observations demonstrates that LaCGKN **performs efficient and accurate Lagrangian DA** and prediction **without reliance on ensembles or the physical model**.
- Developing **digital twins for ENSO** forecasting and DA is ongoing work.

“CGKN: A deep learning framework for modeling complex dynamical systems and efficient data assimilation”, Chen et al. 2025

“Modeling partially observed nonlinear dynamical systems and efficient data assimilation via discrete-time conditional Gaussian Koopman network”, Chen et al. , 2025

“A Lagrangian Conditional Gaussian Koopman Network for Data Assimilation and Prediction” Wang et al. 2026

Questions/suggestions would be appreciated!

But

- As a latent DA method, CGKN transfers the difficulty posed by nonlinear dynamics to encoder-decoder.
- Finding such latent coordinates can be challenging
- The posterior mean μ_z^n is a minimized mean squared error (MMSE) estimator in latent space. However, after pushing it through a nonlinear decoder to the physical space, $\mathcal{D}(\mu_z^n)$ is not the MMSE in general.
- Uncertainty in the latent space also goes through a nonlinear decoder, which can be hard to quantify. We used an auxiliary network to predict the posterior standard error.
- CGKN differs from the standard Koopman theory: CGKN assumes the states can be separated to observed/unobserved parts, even in latent space; Koopman theory does not.
- ...